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torch-harmonics

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Run local tests passing pypi v0.8.0

Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere

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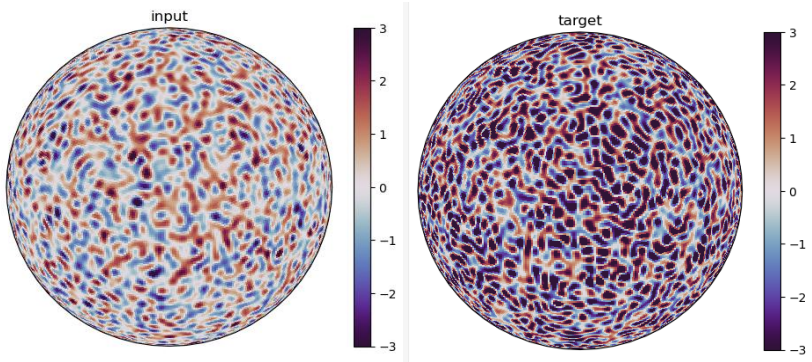


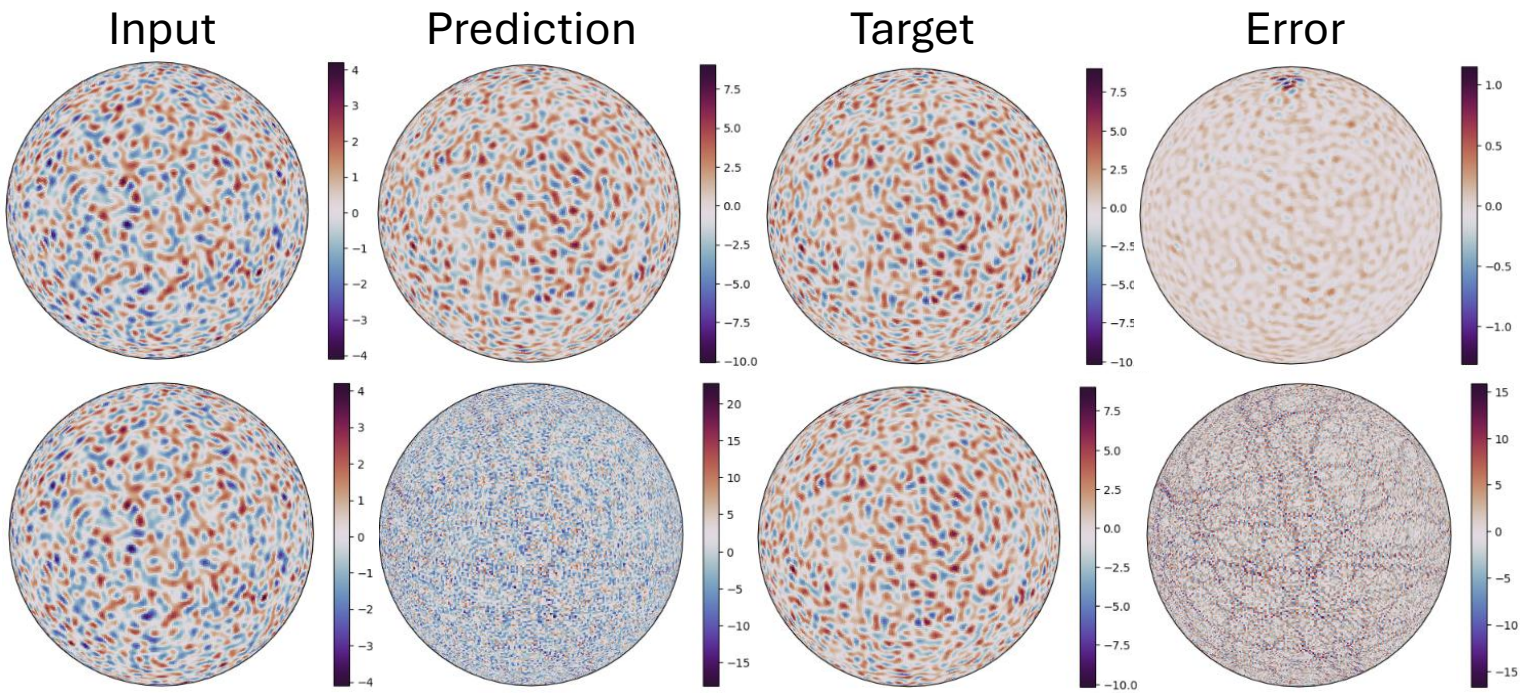
Training Data generated from Shallow Water Equations Solver.

Hyperparameter	Value
Grid size (lat,lon)	241, 480
optimizer, learning rate	Adam, 3e-3
Timestep	30 minutes

(Some of the) Hyperparameters used in Training.

torch-harmonics: Using spherical harmonic transform in place of standard Fast Fourier Transform

Example: Modifying Choice of Loss Function



Top Row: Results with l2 loss (weighs by quadrature weights)

Bottom Row: Results with Spectral L2 loss (same weighing on all spectral frequencies)

Key Takeaway: High frequency coefficients are noisy and distracts operator from learning the lower frequencies that represent physical systems (which majority of large-scale flows are), therefore the poor performance in Spectral L2 loss.